

CBSE ACADEMIC EVALUATION

Class X — Science (Physics, Chemistry & Biology)

Time Allowed: 1 Hour 30 Mins

Subject Code: 086

Maximum Marks: 40

GENERAL INSTRUCTIONS:

1. The question paper comprises four sections: A, B, C, and D. All questions are compulsory.
2. **Section A** contains 4 MCQs carrying 1 mark each.
3. **Section B** contains 3 short answer questions carrying 2 marks each.
4. **Section C** contains 4 short answer questions carrying 3 marks each.
5. **Section D** contains 1 long answer question carrying 5 marks.

SECTION A (Multiple Choice Questions)

4 × 1 = 4 Marks

- Q1.** Which of the following equations represents a balanced chemical neutralization reaction? [1 M]
- (a) $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$ (b) $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$
(c) $\text{Zn} + \text{H}_2\text{SO}_4 \rightarrow \text{ZnSO}_4 + \text{H}_2$ (d) $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
- Q2.** The breakdown of pyruvate to give carbon dioxide, water, and energy takes place in: [1 M]
- (a) Cytoplasm (b) Mitochondria
(c) Chloroplast (d) Nucleus
- Q3.** The focal length of a spherical mirror is +20 cm. The mirror is: [1 M]
- (a) Concave (b) Convex
(c) Plane (d) Concavo-convex
- Q4.** Which part of the human brain controls involuntary actions like blood pressure, salivation, and vomiting? [1 M]
- (a) Cerebrum (b) Cerebellum
(c) Medulla (d) Hypothalamus

SECTION B (Short Answer Questions - Type I)

3 × 2 = 6 Marks

- Q5.** Write the chemical name and formula of Plaster of Paris. State the balanced chemical equation for its preparation from gypsum. [2 M]
- Q6.** Define the term *refractive index*. A ray of light passes from air into a medium with a refractive index of 1.5. Calculate the speed of light in that medium (Given: Speed of light in air = $3 \times 10^8 \text{ m/s}$). [2 M]
- Q7.** Differentiate between aerobic and anaerobic respiration based on the following parameters: [2 M]
1. Requirement of oxygen
 2. End products formed and energy released

SECTION C (Short Answer Questions - Type II)**4 × 3 = 12 Marks**

- Q8.** A circuit contains three resistors of values $2\ \Omega$, $3\ \Omega$, and $6\ \Omega$. [3 M]
1. How should they be connected to obtain a total resistance of $4\ \Omega$? Show the full mathematical calculation.
2. Calculate the total equivalent resistance if all three resistors are connected in parallel.
- Q9.** State Ohm's Law. Draw a neatly labeled circuit diagram to verify Ohm's law showing an ammeter, a voltmeter, a resistor, a plug key, and a battery source. [3 M]
- Q10.** Explain the process of digestion of proteins in the human stomach and small intestine. Name the specific enzymes involved in each stage. [3 M]
- Q11.** Give scientific reasons for each of the following observations: [3 M]
1. Aluminum foil does not react readily with water or steam.
2. Ionic compounds exhibit high melting and boiling points.
3. Metals are good conductors of electricity while non-metals generally are not.

SECTION D (Long Answer Question)**1 × 5 = 5 Marks**

- Q12.** Answer the following questions: [5 M]
1. Draw a neat ray diagram showing the formation of an image by a concave mirror when the object is placed between the focus (F) and the pole (P). State two distinct characteristics of the image formed.
2. A concave lens has a focal length of 15 cm . At what distance should the object from the lens be placed so that it forms an image at 10 cm from the lens? Use the lens formula to calculate the object distance.